

Paper 3: MLP-Based Model for Battery RUL Prediction

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Problem Statement

Predicting the Remaining Useful Life (RUL) of lithium-ion batteries using complex hybrid models often leads to increased computational cost and implementation difficulty. Therefore, there is a need to evaluate whether a simpler neural network model can achieve reasonable accuracy while maintaining low computational complexity. This study focuses on determining the effectiveness of a Multilayer Perceptron (MLP) in predicting battery degradation.

Methodology

The proposed approach utilizes a feedforward neural network (MLP) combined with preprocessing techniques to model battery degradation:

- **Data Preprocessing**
 - Raw battery data is cleaned and normalized.
 - Noise and inconsistencies are reduced to improve data quality.
- **Sliding Window Technique**
 - Time-series data is transformed into supervised learning format.
 - Sequential input-output pairs are generated for model training.
- **MLP Model**
 - A multilayer feedforward neural network is trained on processed data.
 - Learns nonlinear relationships between capacity degradation and cycle count.
- **Prediction Process**
 - The model predicts future battery capacity values.
 - Remaining Useful Life is estimated based on predicted degradation trends.

Key Results

- Achieves Mean Absolute Error (MAE) of approximately 8.52%.
- Root Mean Square Error (RMSE) of approximately 9.59%.
- Performs better than traditional regression-based approaches.
- Demonstrates simplicity and ease of implementation.

Research Gaps / Limitations

- Limited ability to handle noisy and highly irregular data.
- Uses only capacity as input, lacking multi-feature representation.